Perianal Fistula and Abscess: An Imaging Guide for Beginners

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The full digital presentation is available online.

Perianal fistulas are a fairly uncommon entity and are associated with significant morbidity. Diagnostic imaging of the perianal region is key to successful detection and treatment of perianal fistulas. For accurate interpretation and analysis of imaging studies, it is essential to have a comprehensive knowledge of the anal canal and anal sphincter complex anatomy as well as the imaging manifestations of fistulizing disease (Fig 1). This online presentation reviews the normal anatomy at CT and MRI, explains pertinent imaging findings, and describes the pathophysiology of perianal fistulas and the most used classification systems.

Perianal fistula is defined as an abnormal connection between the anal canal and the skin of the perineum. The cause is often idiopathic, but conditions that predispose to the disease include Crohn disease, pelvic infection, tuberculosis, diverticulitis, trauma during childbirth, pelvic malignancies, and radiation therapy. The leading theory of pathogenesis is the cryptoglandular hypothesis, in which the primary event is infection of an intersphincteric gland with formation of an intersphincteric abscess. This can subsequently lead to formation of a fistulous tract.

Imaging of perianal disease is best performed by using cross-sectional techniques, with MRI being the method of choice. MRI allows optimal evaluation of perianal fistulas by providing precise definition of the fistulous tract and any associated complications in the surrounding pelvic structures. T2-weighted MRI sequences without fat saturation are the most useful for depicting the anatomy of the anal canal and anal sphincter complex as they provide excellent soft-tissue contrast. In addition to using the appropriate imaging sequences, it is important for radiologists to evaluate acquisitions in the true axial short-axis and true coronal planes in relation to the anal canal to avoid superimposition of normal structures that can simulate abnormalities.

Treatment of perianal fistulas is predominantly surgical. However, there is a significant incidence of postoperative recurrence and further morbidity. Successful surgical management of perianal fistulas requires a detailed preoperative evaluation of the course of the fistulous tract and the presence of any accessory tracts, abscesses, or other complications. Classification systems have been developed to aid in the treatment of perianal fistulas, with the two main systems being the Parks Classification and the St James's University Hospital Classification (Fig 2). The Parks system describes the course of the primary fistulous tract and its relationship to the anal sphincter complex, while the St James's system incorporates secondary findings that can be seen at MRI.

A practical guide is proposed to facilitate understanding of imaging findings of fistulizing perianal disease. The online presentation reviews the imaging modalities used in assessment of perianal fistulas, with an emphasis on MRI. The pertinent imaging findings of each type of fistula and the most used classification systems are discussed. This is

TEACHING POINTS

- Remember when identifying a perianal fistula that it is an abnormal communication between the anal canal and the perianal or perineal skin from a sinus tract, which is a primary tract that terminates blindly in the subcutaneous tissues.
- Correct anatomic assessment is crucial. Look for the levator ani muscle to define the supralevator space or pelvic cavity, as it is considered a surgical emergency when it is involved.
- Adequate radiologic reporting of perianal fistulas must be clear and concise and should include the following three points: internal opening, course, and external opening.

accomplished with didactic figures that detail the normal anatomy of the anal canal and perianal region, with example cases of the main types of fistulas and their associated complications.

Perianal fistulas can be associated with significant morbidity, and knowledge of how they arise, how to image them, how to describe them, and how to classify them on radiologic images is paramount to ensure appropriate management.

***Disclosures of conflicts of interest.*—K.T.** Board member of the American Board of Radiology.

Suggested Readings

de Miguel Criado J, del Salto LG, Rivas PF, et al. MR imaging evaluation of perianal fistulas: spectrum of imaging features. RadioGraphics 2012;32(1):175–194.
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Khati NJ, Sondel Lewis N, Frazier AA, Obias V, Zeman RK, Hill MC. CT of acute perianal abscesses and infected fistulae: a pictorial essay. Emerg Radiol 2015;22(3):329–335.

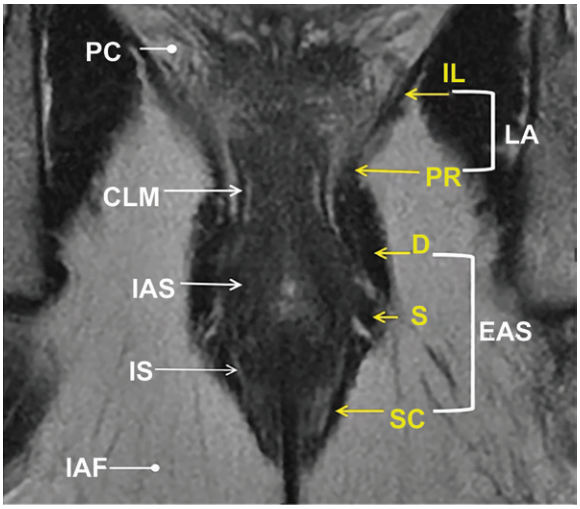

Figure 1. Normal anal anatomy. Oblique coronal T2-weighted MR image demonstrates how the external anal sphincter *(EAS)* contains deep *(D)*, superficial *(S)*, and subcutaneous *(SC)* components, and how the levator ani *(LA)* muscles are composed of iliococcygeus *(IL)* and puborectalis *(PR)* muscles. *CLM* = circular longitudinal muscle, *IAF* = ischioanal fossa, *IAS* = internal anal sphincter, *IS* = intersphincteric space, *PC* = pelvic cavity.

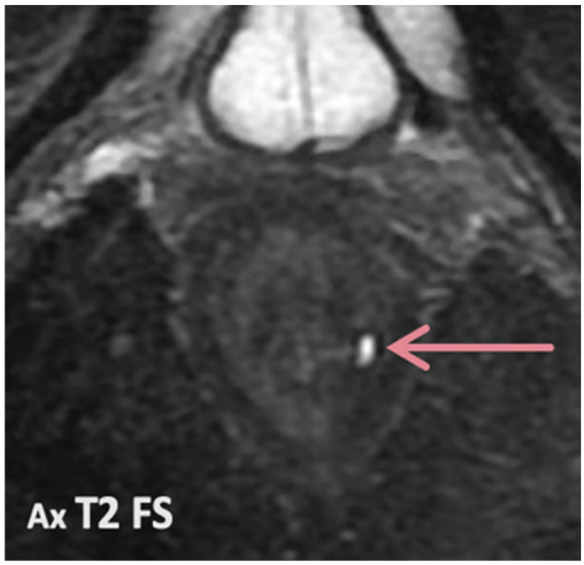

Figure 2. Axial T2-weighted fat-suppressed MR image shows a simple linear intersphincteric fistula tract (arrow) without complications at the 4-o'clock position.